Handwriting Recognition Project Report

Group Members and Division of Labor:

Yaxuan Li: Circuit construction, code writing

Yuetong Li: Code writing, installation production

Zuzhen Jia: Circuit construction, code checking

THE MAIN CONTENT OF THE PROJECT

The main goal of the "Handwriting Recognition" project is to use a Raspberry Pi to recognize handwritten numbers and display them in an eight-segment digital tube. To achieve this, it is divided into three modules – machine learning, recognition, and photo taking.

In the machine learning module, we build our machine learning database by binarizing the training image and splitting it into a two-dimensional array, so that each number is represented by a special array of 0s and 255s.

In the recognition module, we perform the same operation as the previous module on the target image to be recognized, and use the KNN algorithm to compare and fit it with the model in the trained database to achieve the recognition effect. Then the identification result is displayed through the eight-segment digital tube by building a circuit.

In the photo taking modules, we use the external camera and switch of the

Raspberry Pi to realize the switch-controlled camera system, and the photos taken are transmitted to the second module as the image to be recognized.

PRESENTATION OF PROJECT RESULTS

We first placed the digital strip to be identified at the center cross of the bottom plate and cover it with the device, and the quality of the photo was ensured by the light source attached to the device. The camera is fixed to a small hole in the top of the unit to ensure stable focus and to ensure the paper strip is in the center of the lens. Then start the program and press the switch to take a picture of it.

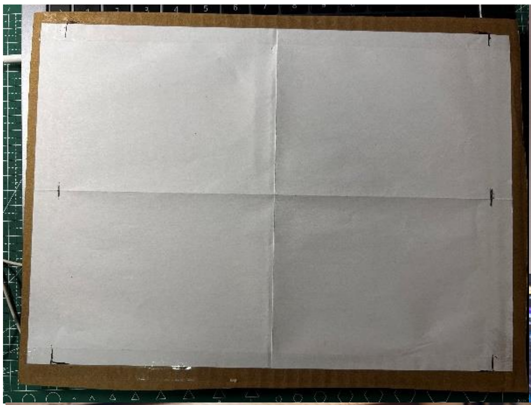
After taking a photo, due to the filter problem of the camera, we applied white balance processing to the photo, and then grayscale and binarization the image, and in the process, we controlled the threshold of binarization to eliminate the noises that may appear in the image. Finally, we cropped the image along the edges of the numbers into an array of rectangular images. In this process, since all the noise may not be dealt with during binarization, we filter out the images with too small pixels from the array when we split the image, so as to ensure the normal segmentation of the image.

Next, we widen the image that is too narrow such as "1", edge all the images and resize them to the same size as the training data, and then inflate the number to achieve a format similar to the training data to improve the recognition rate.

In the final check, we achieved an 80% recognition rate (in fact, 90% in the later adjustment)

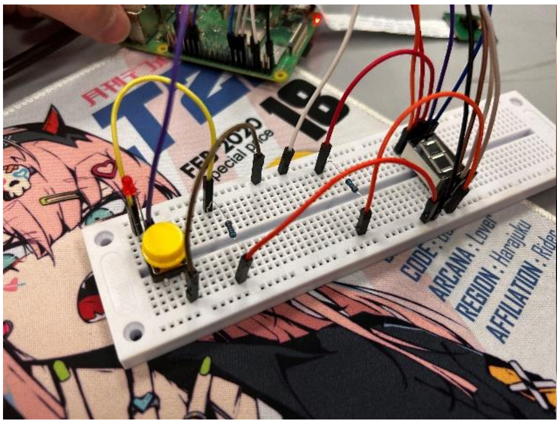
Finally, we were able to correctly display the identified numbers through an eight-segment digital tube.

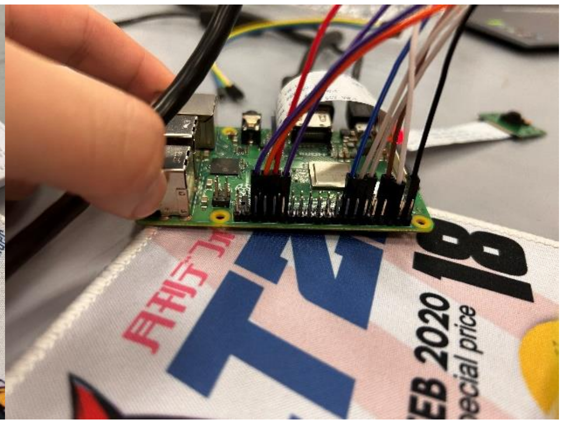
The following is a demonstration of the device, the circuit, and some of the code:



```python
kernel = np.ones((2, 2), np.uint8)

numberList = []
for i in range(0, len(newImgMonos)):
    imgResize = cv2.resize(newImgMonos[i], resizeSize, interpolation=cv2.INTER_AREA)
    img_dilate = cv2.dilate(imgResize, kernel, iterations = 2)
    _threshold, imgBin = cv2.threshold(img_dilate, 10 , 255, cv2.THRESH_BINARY)
    imshow(imgBin)
    imgReshape = img_dilate.reshape(reShapeSize).astype(np.float32)
    _, result, _, _ = knn.findNearest(imgReshape, k=1 )
    numberList.append(int(result))
for j in range(0, len(imgRow)):
    print("The "+str(j)+"th row has:" + str(numberList[j * len(imgMono):(j + 1) * len(imgMono)]))
```





Free Discussion And Summary

We encountered many problems during the completion of this project, but with the help of the teacher assistant and our own exploration, we finally solved these problems one by one, and I selected the following typical problems to elaborate.

Image denoising and segmentation

In the cutting of the target image to be segmented, we found that in practice, shadows are generated due to uneven lighting when taking pictures, resulting in a lot of white noise in the binary image, and if we simply adjust the threshold, it is possible to eliminate the noise, but it will reduce the sharpness of the numbers at the same time, so we only keep the image with a length and width greater than 5 pixels in length and width during cutting. In the later reshape step, the image after reshaping may be distorted for example the image number "1" will be cut too narrow. To solve this, we consider adding black bars about 30 pixels on each side of an image that is less than 50 pixels wide after cutting, and then resize the image to ensure that the numbers are not distorted too much.

Improve recognition rates

The low recognition rate in the implementation of the project has been a problem that has plagued us for a long time, and in order to solve this problem, we initially experimented by adjusting various parameters, but with little success. Later, we compared the training data with the detection data and found that the numbers in the detection data were generally thin, which may lead to inaccurate identification. After consulting with the teacher, we found that the image could be corroded and inflated, so we inflated the cut image, which significantly improved the recognition rate (from 60% to about 90%)

Optimize the quality of the image

In the final practice process, we found that the camera camera photo filter problem has been difficult to solve, there is always vignetting in the photographs that affect the recognition of handwriting, plus the uneven light in the classroom, the shadow of the person will also affect the result after binarization, in order to solve the image problem of taking pictures, we initially tried to adjust the white balance, contrast, brightness and other parameters of the Raspberry Pi camera, but the results are not satisfactory. Later, we chose to manually white-balance the image and successfully corrected it from a cooler color. For the vignetting of the picture, we choose to directly capture the digital part of the picture, so as to avoid the impact of vignetting on the segmentation of the image. In addition, because every time you hold the Raspberry Pi and the circuit board to take a picture, it is easy to produce shadows and cause the network cable connection to be interrupted, and at the same time, the numbers taken at different positions are also in different positions, and the position data of each screenshot is not the same, which is time-consuming and laborious to adjust. So, we found inspiration from the “studio”, made a "black box", fixed the position of the camera which let the picture taken by it can be automatically cut by the program, and the fully wrapped shell and built-in LED lights solve the problem of uneven light and shadows, so that we can save a lot of time in the photo taking moudule for adjusting the parameters. Another point is that after fixing the Raspberry Pi to the circuit board, only one person needs to control the button when taking pictures, which saves a lot of manpower.

FEELINGS

Yuetong Li:

The deepest feeling given to me about completing this project was first and foremost a sense of accomplishment, at least for the first time in my life I worked on an IT project with my teammates, from getting to know the Raspberry Pi, to writing functions for it, to assembling a circuit to implement a series of functions, all of which gave me some excitement to learn something new. When we finally finished the whole project, I felt a sense of joy that we had done it.

The second is to change my view of the entire information technology industry, obviously a very common digital recognition, but the real implementation is far more complicated than imagined, and there are more and more difficult new knowledge waiting for us to master and use, and there are faster and more convenient methods waiting for us to discover. This feeling of "there are people outside the world, and there is heaven outside the sky" made me understand the true meaning of "endless learning".

Yaxuan Li:

Handwriting recognition is the project I was most interested in. In this project, I learned a lot of knowledge about cv2 and machine learning, such as knn algorithm, and also learned many methods and principles of image processing. I participated in the whole process of code writing, so I experienced more setbacks, such as unknown bug reports, and constantly improved the code. For example, when the picture was not clear after zooming, I found that the image could be expanded to improve the clarity and thus improve the recognition rate. In the process of doing the project, my ability to retrieve information on the network was greatly improved. Even if the teacher only explained the principle briefly in class, I could also improve my code by searching the network. All in all, the project of handwriting recognition stimulated my interest in visual direction and made me gain a lot from it.

Zuzhen Jia

Through the Raspberry Pi project, I gained a more detailed understanding of programming and circuit knowledge, as well as the principles of digital recognition. To complete this project, I encountered many difficulties and spent a lot of time and effort searching for information and learning to make our project more successful. Now that the project has achieved some small results, I feel very happy. At the same time, I also feel a little bit of loss because the project has ended and I can no longer be involved in it. However, I know that this is just the beginning, and I have more projects waiting for me to complete and challenge. Overall, completing this Raspberry Pi project has made me feel satisfied and fulfilled, and has also made me look forward to future challenges and achievements.